Korean vocative truncation and Information Theory

Seunghun Lee (ICU) & Shigeto Kawahara (Keio University)

Japanese/Korean Linguistics 24, Oct 14, 2016@NINJAL

Abstract

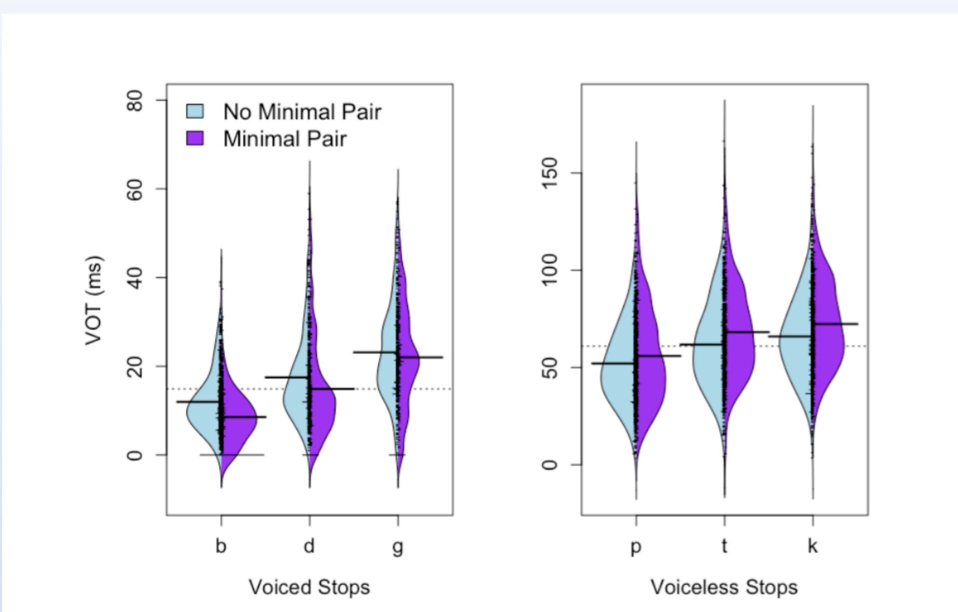
To the extent that language is a system for communication, it would not be surprising if phonology is designed to facilitate efficient communication. A recent approach to phonology, Message-Oriented Phonology (MOP), makes use of Shannon's (1948) Information Theory to formalize this hypothesis (Hall et al. 2016). We think that the evidence that speakers aim for efficient communication is strong in terms of phonetic patterns; but, how "deep" into the linguistic system does the principle of effective communication hold? We maintain that Korean vocative truncation pattern shows that speakers maintain the principle of efficient communication even in a (morpho-)phonological pattern.

Introduction

Message-Oriented Phonology (Hall et al. 2016).

One principle that shapes phonetics/phonology is accurate message transmission.

- Speakers maintain a contrast that is more informative (cf. positional licensing effects).
- In terms of phonetics, speakers implement a more informative contrast more robustly.



Analysis of the Buckeye Corpus (taken from Hall et al 2016): The VOT differences are exaggerated with minimal pair.

This theory itself is inspired by Information Theory proposed by Claude Shannon (1948). The informativity can be measured in terms of Shannon's entropy (to be used in our analysis later).

We think that evidence for MOP is already strong for phonetic patterns. How about (morpho)-phonological patterns? (cf. Shaw et al. 2014 on Chinese truncation, which inspired this project on Korean truncation).

Korean vocative truncation

Basic descriptions

Korean first names consist of two parts: one part is shared by the same generation of siblings and cousins, and the other part is unique to each person. The order between these two elements alternates between one generation to the other.

First generation		Second generation		Third generation		....
hong+hui	[hoŋ.hɨi]	jae+eun	[t͡ʃe.ɨn]	min+su	[min.su]	....
dong+hui	[doŋ.hɨi]	jae+young	[t͡ʃe.jʏŋ]	in+su	[in.su]	
seok+hui	[sʏk.hɨi]	jae+hun	[t͡ʃe.hun]	hui+su	[hi.su]	
yang+hui	[jaŋ.hɨi]	jae+jun	[t͡ʃe.dʒun]	...		
ja+hui	[t͡ʃa.hɨi]	jae+u	[t͡ʃe.u]			

What happens in vocative truncation is to take the unique part of the names, and add [(j)a] to them. [j] appears when the unique part ends with a vowel to avoid hiatus (suggesting that this truncation pattern is governed by linguistic principles).

honga	[hoŋ.a]	euna	[ɨ.na]	mina	[mi.na]
donga	[doŋ.a]	younga	[jʏŋ.a]	ina	[i.na]
seoka	[sʏ.ga]	huna	[hu.na]	huija	[hi.ja]
jaya	[t͡ʃa.ja]	uya	[u.ja]		

Puzzle: many truncation patterns across languages retain initial materials, including Korean (e.g. [haŋkuk] + [ɨnheŋ] -> [hanɨn] `GLOSS') and Japanese ([paasonaru]+[kompuutaa] => [pasokon] `PC') (cf. Beckman 1998; Weeda 1992). What is going on in the vocative truncation pattern?

Analysis

Intuition: Vocative forms are used by the family members. The unique part of the name, rather than the generation marker, is more informative in distinguishing who is being referred to.

Formally: *P*(message| signal, context) is the probability of the listener retrieving the correct message given its signal and context (Hall et al. 2016; Shannon 1948).

Let the context be the hui family. Let us suppose that the intended message is /hong-hui/.

P(hong-hui| [hui], hui) = 1/5 = 0.2 (to generalize 1/n, where n is the # of family members)
P(hong-hui| [hong], hui) = 1 (assuming that there is only one person named hong-hui).

We can also cast these differences in terms of entropy. Given the hui family,

P(hui)=1, its entropy = -log$_2$(1) = 0 bit
P(hong)=0.25, its entropy = -log$_2$(0.25) =2.3 bits

/hong/ has higher *P*(message| signal, context) and higher entropy than /hui/

More on truncation and informativity

When you meet a new person, and do not know which part of the name is the unique portion, truncation is impossible (M-Parse effect: Prince & Smolensky 1993/2004).

Korean university name truncations; [i~~hwajʏ~~-de], [yʏn~~se~~-de], [ko~~rjʏ~~-de]. However, [sʏ~~ul~~+de] is impossible, because [sʏ+de] can either mean [sʏ~~ul~~+de] or [sʏ~~won~~+de].

GENERALIZATION: Korean speakers allow truncation iff *P*(message| signal, context) is (nearly) 1.

Some admittedly sporadic examples that are similar from Japanese (due to research conducted by undergraduate student at ICU).

[suta+kon] <= [misutaa kontesuto]
'Male beauty pageant'
[nai+kara] <= [agenai karaage]
'nondeep-fried fried chicken'

Taking initial portions ([misu] `Ms' or [age] `fried') would miss crucial information.

University name truncation: the dominant pattern is to take the initial kanji (e.g. 東大 from 東京-大学). However, 梨大 from 山梨大学, because 山大 can be 山口大学.

Conclusion

For successful communication, it is important that the speakers' intention is conveyed accurately to the listener.

Formally, *P*(message| signal, context) is kept high.

Korean vocative truncation pattern precisely does this.

It provides support to MOP from a morphophonological perspective (a la Shaw et al. 2014).

Acknowledgements

Thanks to Beth Hume and Jason Shaw for our collaboration. Shaw et al. (2014) was a major inspiration of this work. Thanks also to JSPS grants #26770147, #26770147, #15F15715, which made this project possible.